

Deliverable 2: Relational Design and Data Population

Real Estate Management System

MySQL schema, constraints, indexes, queries, and sample data

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Course	CS241
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Database	MySQL
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1. Team Contributions

The table below records each member's contribution to Deliverable 2.

Team member	Contribution	Output area
Prabhat Singh	Refined the entity-to-relation mapping and checked consistency with the required queries, validate dataset.	Schema design
Pradyumn Ojha	Prepared the MySQL DDL, constraints, triggers, and indexing strategy.	SQL implementation
Vighnesh Mani Tripathi	Generated sample data, improved query coverage, and validated the dataset against the deliverable requirements.	Data population and query support

Relational Schema

Table 1: person

Column	Type	Constraints
person_id	INT	PRIMARY KEY
first_name	VARCHAR(50)	NOT NULL
last_name	VARCHAR(50)	NOT NULL
email	VARCHAR(120)	NOT NULL, UNIQUE

Table 2: person_phone

Column	Type	Constraints
person_id	INT	PRIMARY KEY (composite), FK → person(person_id)
phone_number	VARCHAR(15)	PRIMARY KEY (composite), NOT NULL

Table 3: agent

Column	Type	Constraints
agent_id	INT	PRIMARY KEY
person_id	INT	NOT NULL, UNIQUE, FK → person(person_id)
commission_percent	DECIMAL(5,2)	NOT NULL, CHECK: 0 ≤ value ≤ 100
experience_years	INT	NOT NULL, CHECK: value ≥ 0

Table 4: property

Column	Type	Constraints
property_id	INT	PRIMARY KEY
address	VARCHAR(150)	NOT NULL
city	VARCHAR(60)	NOT NULL
area	VARCHAR(60)	NOT NULL
property_type	VARCHAR(20)	NOT NULL, CHECK: IN ('house', 'apartment')
size_sqft	INT	NOT NULL, CHECK: value > 0
bedrooms	INT	NOT NULL, CHECK: value > 0
year_built	INT	NOT NULL, CHECK: 1900–2100

Table 5: ownership

Column	Type	Constraints
ownership_id	INT	PRIMARY KEY, AUTO_INCREMENT
property_id	INT	NOT NULL, FK → property(property_id)
owner_id	INT	NOT NULL, FK → person(person_id)
start_date	DATE	NOT NULL
end_date	DATE	DEFAULT NULL (NULL = current owner)

Table 6: listing

Column	Type	Constraints
listing_id	INT	PRIMARY KEY
property_id	INT	NOT NULL, FK → property(property_id)
agent_id	INT	NOT NULL, FK → agent(agent_id)
type	VARCHAR(10)	NOT NULL, CHECK: IN ('sale', 'rent')
listed_price	DECIMAL(12,2)	NOT NULL, CHECK: value > 0
list_date	DATE	NOT NULL
status	VARCHAR(20)	NOT NULL, CHECK: IN ('active','sold','rented','withdrawn')

Table 7: transaction_

Column	Type	Constraints
transaction_id	INT	PRIMARY KEY
listing_id	INT	NOT NULL, UNIQUE, FK → listing(listing_id)
customer_id	INT	NOT NULL, FK → person(person_id)
transaction_date	DATE	NOT NULL
final_amount	DECIMAL(12,2)	NOT NULL, CHECK: value > 0

Table 8: rent_detail

Column	Type	Constraints
rent_detail_id	INT	PRIMARY KEY
transaction_id	INT	NOT NULL, UNIQUE, FK → transaction_(transaction_id)
lease_start	DATE	NOT NULL
lease_end	DATE	NOT NULL, CHECK: lease_end > lease_start
monthly_rent	DECIMAL(12,2)	NOT NULL, CHECK: value > 0

2. Design Assumptions

- person is the supertype for individuals in the system, while agent-specific data is stored in agent.
- A property has exactly one current owner at a time, and ownership history is maintained in ownership.
- A listing belongs to one property and one agent, and a listing can produce at most one completed transaction.
- Rental transactions do not change ownership; sale transactions update ownership history through database triggers.
- The table name `transaction_` is used instead of `transaction` to avoid keyword conflicts in SQL.
- For rent transactions, `transaction_.final_amount` stores the agreed monthly rent, while lease details are stored in `rent_detail`.

3. Relational Schema Summary

The implemented schema consists of the following relations.

Relation	Purpose	Important keys or constraints
person	Stores all individuals such as owners, customers, and agents.	Primary key on <code>person_id</code> , unique email.
person_phone	Stores multiple phone numbers for a person.	Composite primary key, foreign key to person.
agent	Stores agent-specific details.	Unique <code>person_id</code> , commission and experience checks.
property	Stores property master data.	Checks on property type, size, bedrooms, and year built.
ownership	Tracks current and historical ownership records.	Foreign keys to property and person.

listing	Stores sale and rent listings handled by agents.	Checks on listing type, status, and price.
transaction_	Stores completed sale and rental transactions.	Unique listing_id, foreign keys to listing and person.
rent_detail	Stores lease information only for rental transactions.	Unique transaction_id, lease-date and monthly-rent checks.

4. Constraints, Triggers, and Indexes

- Check constraints restrict domains such as property type, listing status, positive price values, and valid lease dates.
- Triggers prevent invalid transaction dates, block rent details for sale transactions, and stop listings without a valid current owner.
- After a transaction is inserted, the corresponding listing status is updated automatically to sold or rented.
- Sale transactions also update ownership history automatically by closing the old record and creating a new one.
- Indexes were added on search-heavy paths such as city and area, listing agent and status, listing type and date, and transaction date.

5. Data Population Summary

The sample dataset was expanded so that the required project queries return meaningful results.

Table	Rows inserted	Notes
person	30	People acting as owners, customers, and agent profiles.
person_phone	24	Supports multiple phone numbers for selected persons.
agent	12	Active agents in the system.
property	24	Houses and apartments across Guwahati locations.

ownership	24+	Initial ownership rows plus trigger-generated sale updates after execution.
listing	25	Sale and rent listings with active, sold, and rented statuses.
transaction_	17	Includes 2018 and 2023 sale coverage and 2024 rental transactions.
rent_detail	6	Lease details for completed rental transactions.

- The dataset includes houses built after 2023 that are available for rent.
- It includes an active GS Road rent listing with at least two bedrooms and rent below Rs. 15,000.
- It includes 2023 sale transactions with a clear top-selling agent by count and total amount.
- It includes enough 2018 sales to support average selling price and average time-onmarket analysis.

6. Query Coverage

The accompanying query script covers the required deliverable questions and one additional reporting query.

- Houses built after 2023 that are available for rent.
- Average selling price of properties sold in 2018.
- Active rent listings on GS Road with at least two bedrooms and monthly rent below Rs. 15,000.
- Agent who sold the most properties in 2023, with total sale amount used as the tiebreaker.
- Average time properties remained on the market for each agent in 2018.
- Details of the most expensive house and the house with the highest rent in the database.

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- Tool used: ChatGPT.

7. Submission Files

- CS241_Deliverable_2_Schema_Data.sql - executable MySQL script containing schema, constraints, triggers, indexes, and sample data.
- CS241_Deliverable_2_Queries.sql - MySQL script containing the required project queries.
- CS241_Deliverable_2_Report.pdf - formatted Deliverable 2 report for submission.

8. AI Usage Note

- Used for: Generating realistic data to populate the database and help in making triggers, index and adjusting data according to that.